Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing with Business Management**

Assessment	**Algorithms and Mathematical Concepts for Computing**
Assessment Type	**Examination**
Date	**Wednesday 28th May 2025**
Time	**2.00pm – 5.00pm**

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

This question relates to the Software Development Process and Algorithm:

a) During the early stages of the software development lifecycle, a phase focuses on understanding the problem domain, identifying stakeholder needs and defining what the system should achieve. What is the name of this phase? **1**

Mark scheme

Requirements Analysis (1 mark)

b) Which stage of the software development lifecycle focuses on addressing bugs, implementing improvements, and providing ongoing support for the system? **1**

Mark scheme

Maintenance (1 mark)

c) i) Define a flowchart **and** explain its purpose in software development. **2**

ii) Compare flowcharts with pseudocode. **2**

iii) Write pseudocode to display a greeting message five times using a loop. Use selection so that the message for the first time (i.e. when the counter is 1) is different from the rest (e.g., display “Welcome!” the first time and “Hello again!” for the remaining times). After the greetings have been displayed five times, display a final message such as “Session over.” **4**

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Mark scheme

i) ***Definition of Flowchart:*** *A flowchart is a graphical representation of an algorithm or process using symbols and arrows to illustrate the flow of execution.* ***(1 mark)***

Purpose of Flowchart: *It helps visualise the logic of a program and improves understanding before implementation.* ***(1 mark)***

ii) ***Comparison with Pseudocode:***
Flowcharts: Visual and easy to understand, useful for explaining processes to non-programmers. ***(1 mark)***

Pseudocode: Uses structured text to describe an algorithm, making it easier to translate into actual code. ***(1 mark)***

iii) ***Example of pseudocode to display a message.:***

Pseudocode:

```
SET counter = 1
WHILE counter <= 5
   IF counter = 1 THEN
      DISPLAY "Welcome!"
   ELSE
      DISPLAY "Hello again!"
   END IF
   INCREMENT counter by 1
END WHILE

DISPLAY "Session over"
```

Correct Use of Loop (1 mark):

- *The pseudocode includes a loop (e.g., WHILE or FOR) that repeats the greeting five times based on the counter.*

Proper Use of Selection for Different Messages (1 mark):

- *The pseudocode correctly includes a selection (e.g., IF statement) to display a different message for the first iteration ("Welcome!") and another message for the rest ("Hello again!").*

Accurate Increment of Counter (1 mark):

- *The counter is correctly incremented after each iteration (e.g., INCREMENT counter by 1), ensuring the loop ends after 5 iterations.*

Final Message Displayed After 5 Iterations (1 mark):

- *After the loop finishes, the pseudocode displays a final message (e.g., "Session over").*

Total 10 Marks

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Question 2

This question relates to numbers and number systems:

a) Which of the following are true about number systems? **2**

a) In a binary system, the digits are limited to the values 0 and 1.
b) The decimal system uses base 8.
c) The hexadecimal system is often used in computer memory addressing because it represents values in base 10.
d) In the octal system, each digit can range from 0 to 7.

Mark scheme

a) In a binary system, the digits are limited to the values 0 and 1. (1 mark)
d) In the octal system, each digit can range from 0 to 7. (1 mark)

b) Define the greatest common divisor (GCD) **and** explain its significance in number theory. **2**

Mark scheme

- ***GCD is the largest positive integer that divides two or more numbers without a remainder. (1 mark)***
- ***It is significant in simplifying fractions and solving problems related to divisibility and modular arithmetic. (1 mark)***

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c) Perform the following conversions **and** show your working. **2**

i) Convert the decimal number 91 to binary. Show your working.

Mark Scheme

- ***91 ÷ 2 = 45 remainder 1***
- ***45 ÷ 2 = 22 remainder 1***
- ***22 ÷ 2 = 11 remainder 0***
- ***11 ÷ 2 = 5 remainder 1***
- ***5 ÷ 2 = 2 remainder 1***
- ***2 ÷ 2 = 1 remainder 0***
- ***1 ÷ 2 = 0 remainder 1***
- ***Now, read the remainders from bottom to top: 1011011***

- ***Correct answer: 1011011***
  (1 mark for correct answer, 1 mark for working)

ii) Convert the binary number 1101 to octal. Show your working. **2**

Mark Scheme

The binary number is 1101.

Grouping from the right: 1 101 → Pad the first group with two zeros: 001 101

Now convert each group to octal:

001 → 1

101 → 5

Thus, 1101 in binary is 15 in octal.

Correct answer: 15
(1 mark for correct answer, 1 mark for working)

iii) Convert the hexadecimal number 2F to decimal. Show your working. **2**

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Mark Scheme

The hexadecimal number is 2F.

The first digit is 2, and the second digit is F (which is 15 in decimal).

Now, calculate the decimal value:

(2 × 16^1) + (15 × 16^0)

= (2 × 16) + (15 × 1)

= 32 + 15

= 47

Correct answer: 47

(1 mark for correct answer, 1 mark for working)

Total 10 Marks

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Question 3

Answer the following questions related to logic and Boolean algebra:

a) Which of the following symbols represents the logical NOT operation? **1**

i) $\wedge$ **ii)** $\neg$
iii) $\rightarrow$ **iv)** $\vee$

Mark scheme

ii) ¬ (1 mark)

b) Complete the truth table for the Boolean expression: **4**

$$(P \wedge Q) \vee (\neg P \wedge Q)$$

P	**Q**	**P ∧ Q**	**¬P**	**(¬P ∧ Q)**	**(P ∧ Q) ∨ (¬P ∧ Q)**
F	F				
F	T				
T	F				
T	T				

Mark scheme

Accurate completion of the truth table (each correct column = 1 Marks)

Please note *that if the result of the third column (P ∨ Q) and/or the fourth column* **¬P** *is incorrect, but the calculation for the fifth column* **(¬P ∧ Q) and/or sixth column (P ∧ Q) ∨ (¬P ∧ Q)***is correct based on the incorrect results of the third and/or fourth column, the mark will still be awarded.*

Answer:

P	**Q**	**P ∧ Q**	**¬P**	**(¬P ∧ Q)**	**(P ∧ Q) ∨ (¬P ∧ Q)**
F	F	F	T	F	F
F	T	F	T	T	T
T	F	F	F	F	F
T	T	T	F	F	T

c) Write and explain the concept of the First De Morgan's Theorem **and** provide ONE (1) example. Include the truth table for the theorem. **5**

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Mark Scheme

- ***First Theorem: ¬(P ∧ Q) = ¬P ∨ ¬Q***

  1 mark for correctly defining the first theorem

- ***Explanation: The first De Morgan's theorem states that the negation of an AND operation is equivalent to the OR of the negations of the individual terms. It simplifies Boolean expressions involving NOT and AND operations.***

  1 mark for explaining the theorem

- ***Example 1: "The statement 'It is not the case that both the lights are on and the door is open' is equivalent to saying that either the lights are off or the door is closed."***

  This demonstrates the first theorem, where the negation of an AND operation is equivalent to the OR of the negations of the individual terms. (1 mark)

- ***Truth table for the First Theorem:***

P	***Q***	***(¬(P ∧ Q) = ¬P ∨ ¬Q)***
T	***T***	***F***
T	***F***	***T***
F	***T***	***T***
F	***F***	***T***

(Total of 2 marks: 1 mark for two correct rows.)

Total 10 Marks

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Question 4

Answer the following questions related to Set Theory:

a) Which of the following represents the Cartesian Product of two sets? **1**

i) $A \cap B$ **ii)** $A \times B$

iii) $A \cup B$ **iv)** $A \subset B$

Mark scheme

Answer: ii) $A \times B$ ***(1 mark)***

b) Define the concept of a subset **and** provide an example. **2**

Mark Scheme

- ***A subset is a set where every element of set A is also in set B. (1 mark)***
- ***Example:***
  If A = {1, 2} and B = {1, 2, 3, 4}, then $A \subseteq B$. (1 mark) Accept valid alternatives.

c) Find the complement of the set B = {2, 4, 6} relative to the universal set U = {1,2,3,4,5,6,7} **and** explain the process. **2**

Mark Scheme

- ***Correct complement: $B' = \{1, 3, 5, 7\}$.***
  (1 mark for correctly identifying the complement of B)
- ***The complement of a set includes all the elements in the universal set U that are not in B. Therefore, $B' = \{1, 3, 5, 7\}$.***

(1 mark for explaining the process)

d) Define the term "intersection of sets" and find the intersection of A = {1, 2, 3}, B = {2, 3, 4}, and C = {3, 4, 5}. Show the intersections of two sets at a time. **3**

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Mark Scheme

- ***The intersection of two sets, denoted as A ∩ B, is the set of elements that are common to both A and B.***
  (1 mark for explaining the intersection)

Calculation

- ***Step 1: Find the intersection of sets A and B.***
  - ***A = {1, 2, 3}, B = {2, 3, 4}***
  - ***A ∩ B = {2, 3}***

  (1 mark)
- ***Step 2: Find the intersection of the result from Step 1 with set C.***
  - ***A ∩ B = {2, 3}, C = {3, 4, 5}***
  - ***(A ∩ B) ∩ C = {3}***

  (1 mark)

e) Explain the difference between a finite set and an infinite set with an example for each.

2

Mark Scheme

- ***Finite Set: A set with a countable number of elements***

  Example: {1, 2, 3, 4} (1 mark)
- ***Infinite Set: A set with uncountable elements***

  Example: Set of natural numbers {1, 2, 3, …} (1 mark)

Total 10 Marks

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Question 5

Answer the following questions related to statistics:

a) Explain the concept of standard deviation. Provide **ONE (1)** example where it is useful in data analysis. **2**

Mark Scheme

- ***Standard deviation measures the dispersion or spread of data points around the mean. A higher standard deviation indicates greater variability, while a lower standard deviation suggests data points are closer to the mean. (1 mark)***
- ***Example: Standard deviation is useful in finance to measure stock market volatility. A high standard deviation suggests a risky stock with large price fluctuations. (1 mark)***

b) Write the formula for standard deviation **and** calculate the standard deviation for the dataset: 3, 7, 5, 9, 6. Show your working. **3**

Mark Scheme

1. ***Formula: $\sigma = \text{sqrt}(\Sigma(xi - \text{mean})^2 / n)$ (1 mark)***
2. ***Calculate the mean:***
   Mean = (3+7+5+9+6) / 5 = 6
   (1 mark)
3. ***Compute standard deviation:***

   $\sigma = \text{sqrt}(((3-6)^2 + (7-6)^2 + (5-6)^2 + (9-6)^2 + (6-6)^2) / 5)$

   $= \text{sqrt}((9+1+1+9+0)/5)$

   $= \text{sqrt}(20/5)$

   $= \text{sqrt}(4)$

   $= 2$

   (1 mark)

c) Define the term "skewness" in statistics. Explain the difference between positive **and** negative skewness. **2**

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Mark Scheme

- ***Skewness measures the asymmetry of a dataset's distribution. (1 mark)***
- ***Difference:***
    - ***Positive skew: The right tail is longer, meaning the majority of values are concentrated on the left.***
    - ***Negative skew: The left tail is longer, meaning the majority of values are concentrated on the right.***

      (1 mark)

d) Define mode. **3**
Find the mode of the dataset: 2, 3, 3, 5, 7, 7, 7, 8, 9.
Provide an example of when the mode is useful.
Mark Scheme

1. ***Definition: Mode is the most frequently occurring value in a dataset. (1 mark)***
2. ***Identify the mode: The mode is 7, as it appears most frequently. (1 mark)***
3. ***Explanation: The mode is useful in identifying common trends or preferences in data, such as popular product sales. (1 mark)***

Total 10 Marks

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Question 6

Answer the following questions related to Data Structures and Arrays:

a) Which of the following operations is **NOT** typically supported by a stack? **1**

i) Push
ii) Pop
iii) Peek
iv) Dequeue

Mark scheme

iv) Dequeue ***(1 mark)***

b) Define a "linked list," describe **TWO (2)** primary operations performed on a linked list **and** provide an example application of a linked list. **4**

Mark Scheme

Definition: A linked list is a linear data structure where elements (nodes) are stored in a sequence, each containing data and a reference (or link) to the next node in the list. (1 mark)

- ***Operations:***

  Insert operation: Adds a new node with data at a specific position in the linked list. (1 mark)

  Delete operation: Removes a node from the linked list at a specific position. (1 mark)

- ***Application of linked list: Linked lists are commonly used in applications such as implementing dynamic memory allocation or managing playlists in media players, where the size of the list can change dynamically, and quick insertion/removal of elements is needed. (1 mark: The answer is expected to provide just one application. Any of the given applications or any other valid application will be accepted.)***

c) Write a function in pseudocode to insert a node at the beginning of a singly linked list. **5**
You can assume the existence of the function createNode(x) which creates a new node containing data (x) , returning a pointer to the node.

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Mark scheme

Function insertAtBeginning(linkedList, data): # (1 mark) for correct function structure - Function and End Function, # (1 mark) for Parameters

newNode = createNode(data) # (1 mark) Create a new node

newNode.next = linkedList.head # (1 mark) Set new node's next to current head

linkedList.head = newNode # (1 mark) Set new node as the head of the list
End Function

Total 10 Marks

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Question 7
Answer the following questions related to Data Structures:

a) What is the maximum number of elements that can be stored in an array of size *n* at a given index? **1**

i) n **ii)** n/2

iii) 1 **iv)** $2^{(n+1)}$

Mark scheme

iii) 1 ***(1 mark)***

b) Define the term "linked list" and explain its key property. **2**

Mark Scheme

- ***A linked list is a linear data structure where each element (node) contains data and a reference (or link) to the next node in the sequence. (1 mark)***
- ***Key property: Linked lists allow dynamic memory allocation and efficient insertions/deletions at both ends. (1 mark)***

c) Explain the difference between a stack and a queue in terms of data access order. Mention typical use cases for each. **3**

Mark Scheme

- ***A stack follows the Last In First Out (LIFO) principle, while a queue follows the First In First Out (FIFO) principle. (1 mark)***
- ***Typical use cases:***

  Stacks are used for undo operations, recursion, and function calls. (1 mark)
  Queues are used for managing tasks like print jobs, task scheduling, and handling requests. (1 mark)

d) Write pseudocode to push an element onto a stack. Complete the function body below. Make sure the case where the stack is full is handled. **4**

```
Function push(stack, value):
    [BODY]
End Function
```


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Mark Scheme

```
Function push(stack, value):
    if stack.top < stack.capacity:  # (1 mark) Check if there is space in the
stack
        stack[stack.top] = value  # (1 mark) Add the new element to the top
of the stack
        stack.top = stack.top + 1  # (1 mark) Update the top index
    else:
        print("Stack is full")  # (1 mark) Handle the case where the stack is
full
End Function
```

Total 10 Marks

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Question 8

Answer the following questions related to algorithm efficiency and searching techniques:

a) Which of the following sorting algorithms has the worst-case time complexity of $O(n^2)$? **1**

i) Quick Sort **ii)** Merge Sort

iii) Bubble Sort **iv)** Heap Sort

Mark scheme

***iii) Bubble Sort* (1 mark)**

b) Describe the concept of 'divide and conquer' as used in sorting algorithms, with an example of a sorting algorithm that uses 'divide and conquer'. **2**

Mark scheme

- ***Divide and conquer splits a problem into smaller subproblems, solves each one independently, and then combines the results. (1 mark)***
- ***Example: Merge Sort (recursively) divides the array and then merges sorted subarrays. (1 mark)***

c) Write pseudocode for the Bubble Sort algorithm. Complete the body of the function below: **4**

```
Function bubbleSort(arr):
    [BODY]
End Function
```

Mark scheme

```
Function bubbleSort(arr):
    for i = 0 to length(arr) - 1:  # (1 mark) Outer loop
        for j = 0 to length(arr) - i - 1:  # (1 mark) Inner loop
            if arr[j] > arr[j+1]:  # (1 mark) Compare
                swap(arr[j], arr[j+1]) # (1 mark) swap
End Function
```

d) Explain the difference between Merge Sort and Quick Sort in terms of their time complexities. Which data structure is more suitable for each algorithm.? **3**

Mark Scheme

- ***Merge Sort: Has a consistent time complexity of O(n log n) in all cases. (1 mark)***
- ***Quick Sort: Has an average-case time complexity of O(n log n) and a worst-case of $O(n^2)$. (1 mark)***
- ***Merge Sort is preferred for linked lists, while Quick Sort is more efficient for arrays. (1 mark)***

Total 10 Marks

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Question 9

Answer the following questions related to sorting algorithms and recursion:

a) Which of the following sorting algorithms repeatedly compares adjacent elements and swaps them if they are in the wrong order, progressively moving elements toward their correct positions? **1**

i) Bubble Sort
ii) Quick Sort
iii) Selection Sort
iv) Merge Sort

Mark Scheme

iii) Bubble Sort **(1 mark)**

b) Explain selection sort and bubble sort in terms of how they compare and swap elements. After explaining, outline the key difference between the operations of selection sort and bubble sort regarding comparing and swapping elements in one sentence. **3**

Mark Scheme

- ***Selection Sort Explanation: In selection sort, the algorithm selects the smallest (or largest) element from the unsorted part of the list and swaps it with the first unsorted element. (1 mark)***
- ***Bubble Sort Explanation: In bubble sort, the algorithm compares adjacent elements and swaps them if they are in the wrong order, "bubbling" the largest (or smallest) element to its correct position after each pass. (1 mark)***
- ***Comparison of Operations: Selection sort makes fewer swaps (one per pass), while bubble sort may make multiple swaps during each pass. (1 mark)***

c) Fill in the blanks in the pseudocode for computing the nth Fibonacci number using recursion: **4**

```
Function fibonacci(n):
  if n <= _____:
     return _____
  return _____ + _____ (n - 2)
End function
```


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Mark scheme

```
Function fibonacci(n):
  if n <= 1:
     return n
  return fibonacci(n - 1) + fibonacci(n - 2)
End function
```

Mark Scheme:

- ***1 mark for correctly completing the base case check: if n <= 1:***
- ***1 mark for correctly returning the value when the base case is met: return n***
- ***1 mark for correctly handling the recursive addition: return fibonacci(n - 1) +***
- ***1 mark for correctly completing the second recursive call: fibonacci(n - 2)***

d) Explain how recursion can be used to calculate the factorial of a number. **2**

Mark scheme

- ***Recursion is used to break the problem of calculating factorial(n) into smaller subproblems by computing factorial(n-1) and multiplying it by n. (1 mark)***
- ***The base case is when n equals 1, where the recursion stops and returns 1, starting the process of unwinding. (1 mark)***

Total 10 Marks

Question 10

Answer the following questions related to tree data structures:

a) Which of the following is **TRUE** about a binary tree? **1**

i) A binary tree always has exactly two children per node.

ii) The root of the binary tree must have the smallest value.

iii) A binary tree can have any number of children per node.

iv) A binary tree can have at most two children per node, but not necessarily both.

Mark scheme

- ***iv) A binary tree can have at most two children per node, but not necessarily both. (1 mark)***

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b) In a binary search tree (BST), for any given node, all the values in its _____ subtree are smaller, and all the values in its _____ subtree are greater. **1**

Mark scheme

"left, right" (1 mark)

c) Construct a binary tree for the following sequence of numbers: 10, 5, 15, 2, 8, 12, 20. Ensure that 10 is the root. **4**

Mark scheme

```
          10          (1 mark)
        /    \
      5        15     (1 mark)
    /   \    /   \
   2    8   12   20   (1 mark for 2 and 8, 1 mark for 12 and 20)
```

d) Using the binary tree constructed in Question 10c, explain the steps to search for the value 8. Describe each comparison and movement until the value is found or determined to be absent. **4**

Mark Scheme

1. ***Start at the root node (10). (1 mark)***
2. ***Compare 8 with 10: 8 < 10, so move to the left subtree (5). (1 mark)***
3. ***Compare 8 with 5: 8 > 5, so move to the right subtree (8). (1 mark)***
4. ***Compare 8 with 8: 8 == 8, so the value is found. (1 mark)***

Please note that if the answer for the tree constructed in c) is incorrect, but the explanation for searching 8 is correct relative to the constructed tree in c), the student will still receive full marks.

Total 10 Marks

End of paper

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Learning Outcomes matrix

Task	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	LO1	Yes
2	LO5	Yes
3	LO2, LO3	Yes
4	LO4, LO6	Yes
5	LO9	Yes
6	LO7	Yes
7	LO7	Yes
8	LO8	Yes
9	LO8	Yes
10	LO7, LO8	Yes

Grade descriptors.

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Explain and use Computational Thinking	Demonstrates no understanding of Computational Thinking and its role.	Provides a basic explanation of Computational Thinking.	Discusses specific examples of Computational Thinking in various contexts.	Applies Computational Thinking to solve complex problems effectively.	Provides critical analysis and innovative approaches to Computational Thinking.
Explain and apply Boolean algebra concepts and notation.	Unable to explain Boolean algebra concepts and notation.	Provides a basic explanation of Boolean algebra concepts and notation.	Draws logic diagrams and writes Boolean expressions accurately.	Applies Boolean algebra concepts proficiently in various computing scenarios.	Offers critical analysis and optimizations of Boolean algebra operations.
Explain and use propositional logic concepts and notation	Demonstrates no understanding of propositional logic concepts and notation.	Provides a basic explanation of propositional logic concepts and notation.	Uses truth tables proficiently to evaluate propositions.	Applies propositional logic effectively in analysing complex statements.	Offers critical evaluation of propositional logic limitations and implications.
Explain and use simple predicate logic statements	Unable to explain simple predicate logic statements.	Provides a basic explanation of simple predicate logic statements.	Writes and interprets predicate logic statements involving quantifiers accurately.	Applies predicate logic statements effectively in real-world scenarios.	Provides innovative applications and thorough analysis of predicate logic statements.
Understand and explain number based system	Unable to understand or explain number-based systems.	Provides a basic explanation of number-based systems.	Performs conversions and arithmetic operations accurately in different number bases.	Applies number-based systems proficiently in solving computational problems.	Offers advanced insights and applications of number-based systems.

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Explain and use set theory and notation	Unable to explain set theory concepts and notation.	Provides a basic explanation of set theory and notation.	Uses set notation proficiently to perform operations and interpret Venn diagrams.	Applies set theory concepts effectively in analysing complex relationships.	Offers advanced interpretations and applications of set theory concepts.
Identify and use common data structures	Demonstrates no understanding of common data structures.	Provides a basic explanation of common data structures.	Applies common data structures proficiently in solving computational problems.	Demonstrates advanced understanding and optimization of data structure usage.	Offers innovative applications and optimizations of common data structures.
Explain and use common algorithms	Unable to explain common algorithms for searching & sorting.	Provides a basic explanation of common algorithms for searching & sorting.	Explains efficiency criteria used to evaluate algorithms effectively.	Applies common algorithms proficiently in solving computational problems.	Offers critical analysis and optimizations of algorithm efficiency.
Understand the fundamental concepts on statistics	Unable to understand or explain key statistical measures and concepts.	Provides a basic explanation of key statistical measures and concepts.	Calculates and uses descriptive statistics effectively.	Applies statistical concepts proficiently in analysing real-world data.	Offers advanced insights and applications of statistical concepts

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